

What do we mean by
Hyper-Local Economies?

Finding regenerative and
restorative supply chains
within our region, circulating
resources and money
locally, and reducing the
span and impact of our
environmental footprint.

Land Acknowledgement:

This project takes place
on Mohican land, between
the Mahicannituck (river
that flows two ways) and
Taghkanic mountain range.

Hyper-Local Robotic Entanglements
is a project led by FSD.TEMP
that explores the design and
implementation of a DIY backyard
robot to build with natural materials
gathered from local sources. HLRE
explores how to encourage the
tangibility of emerging technologies
through public engagement and
physical artifacts.

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Hyper-Local Robotic Entanglements is made possible by funding
by **Independent Projects**, a grant program of the New York State
Council on the Arts and The Architectural League of New York.
Independent Projects is made possible by public funds from the
New York State Council on the Arts, with the support of the Office
of the Governor and the New York State legislature; and from the
Capital Region Arts, Thrive, & Grow Grant. Arts, Thrive, & Grow
is funded by New York State & the NYSCA, with support from
Governor Kathy Hochul, Senate Majority Leader Andrew Stewart
Cousins, and local elected officials, including Senators Jake
Ashby and Patricia Fahy and Assemblymembers Scott H. Bendett,
John T. McDonald III, Angelo Santabarbara, Phil Steck, and Mary
Beth Walsh; and by the generosity of space, time, and community
from **NATURElab and the Sanctuary for Independent Media** in
North-Central Troy, NY.

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Phragmites Australis: abundant plant matter



Exploring local supply chains through
ecological availability.

Hyper-Local
Robotic Entanglements
FSD.TEMP

2025-2026



Material Economies

Phragmites as Straw Replacement

What is abundant in our local ecosystem? What will the impact be if we harvest it?

Phragmites Australis is non-native, seasonally abundant grass reed that grows densely, outcompeting marsh plants that are significant contributors to our local ecosystem. We chose this material to be the straw in our mudbrick, providing additional shear strength and fiber reinforcement. Rich in cellulose, Phragmites has been explored as a base material in fiber creation, paper making, and weaving.



Responsible harvesting

When working with a plant that spreads quickly, it is important to practice strategies that prevent unintentional spread. In the case of Phragmites, cut seed plumes into garbage bags so they don't scatter and cut reeds underwater to prevent regrowth. Phragmites will often grow amongst native plants when overtaking a marshy area; take care to avoid disturbing beneficial species. *When it doubt, leave it be.*

Identifying material potential in a plant resource
Observe the potential material. Does it remind you of an existing material?

Physical Scale and Components - Size, weight, leaf, stem, juice or milk, fluff, seed, plume, etc.
Ecological Scale - How much is available when harvesting responsibly?

Physical Properties - Does it bend, break, smush? Is there fiber? Does this change when wet, dry, fermented? **Could you use an existing method to work with it?** - Weaving, mixing, felting, etc.



FIG. 90. *Phragmites vulgaris* (Lam.) B. S. P. (*P. communis* Thun.) COMMON REED.—a, A spikelet; b, the outer empty glumes; c, a single flower showing palea, stamens, and stigma.

90. **PHRAGMITES** Thun. Fund. Agrost. 184. 1820. Spikelets loosely 8- to 7-flowered; rachilla articulated above the empty glumes and between the flowers, clothed with long, silky hairs. The lowest flower staminate or neuter, those above hermaphrodite; empty glumes unequal, lanceolate, acute, shorter than the flowers; flowering glumes glabrous, very narrow and long acuminate pointed, much exceeding the short palea. Grain free. Tall, reed-like perennial grasses with stout leaty culms and large terminal panicles.

Species 2 or 3, 1 cosmopolitan, 1 in Asia, and 1 in South America.

